Experiment-5

Write a Servlet to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.

AIM:

To demonstrate the difference between the HTTP GET and POST methods by creating a form in an HTML page and handling the form submissions using two different HTTP methods in a Java Servlet.

ALGORITHM:

1. Input:

- The user fills out a web form with their **name** and **email**.
- The form will use both **GET** and **POST** methods for data submission.

2. Steps for Creating and Handling the HTTP GET and POST Requests:

1. **Create an HTML Form:**
    - Design an HTML form with two input fields: one for the **name** and one for the **email**.
    - Create two buttons to allow the user to submit the form using **GET** or **POST** methods.
    - Use the action attribute to define the **Servlet URL** and specify the form method (either GET or POST).
2. **Create the Servlet to Handle GET and POST Requests:**
    - Write a Java Servlet class that will handle **both GET and POST requests**.
    - The servlet will have two methods:
        - doGet() to handle the GET requests.
        - doPost() to handle the POST requests.
    - In both methods, retrieve the form data using request.getParameter().
    - In the doGet() method, process and display the data in the URL.
    - In the doPost() method, process and display the data in the HTTP request body.
3. **Handle GET Requests in the Servlet:**
    - In the doGet() method, use request.getParameter() to retrieve the name and email submitted via GET.

- Output the data as a response, including the **name** and **email** in the URL (visible to the user).

4. **Handle POST Requests in the Servlet:**
   - In the doPost() method, use request.getParameter() to retrieve the name and email submitted via POST.
   - Output the data as a response, displaying the **name** and **email** without exposing them in the URL (more secure).
5. **Return the Output:**
   - The Servlet should return an HTML page displaying the **name** and **email** entered by the user.
   - For the **GET method**, the output should be visible in the URL.
   - For the **POST method**, the output should be visible in the response body, not the URL.
6. **Deploy and Test the Servlet:**
   - Compile the Servlet class and deploy it to a **Servlet container** (like **Apache Tomcat**).
   - Test the form using **GET** and **POST** methods to see the difference in data submission.
7. **Observe and Compare GET and POST Results:**
   - Test the **GET method**: The form data is sent in the URL.
   - Test the **POST method**: The form data is sent in the HTTP request body, not visible in the URL.

PROGRAM:

Demo.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Web Form</title>
<style>
body{
	font-family: Arial;
	background-color: grey;
}
.container{
	width: 400px;
	margin:auto;
	margin-top: 100px;
	padding:20px;
```

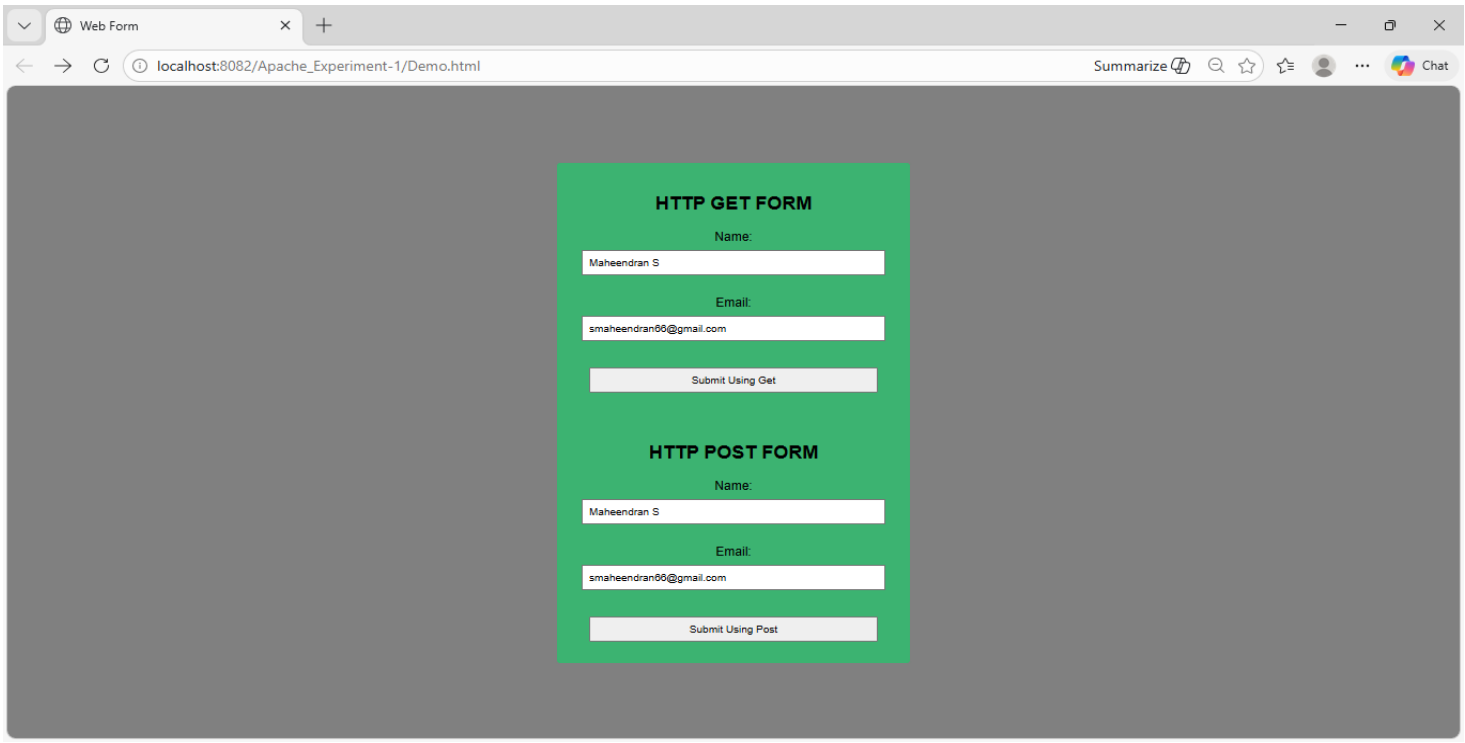
```
        background: mediumseagreen;
        border: 1px solid gray;
        border-radius: 5px;
        text-align: center;
}
input{
        width:90%;
        padding: 8px;
        margin: 8px 0;
}
button{
        padding: 8px 15px;
        margin-top: 10px;
        background-color: grey;
}
</style>
</head>
<body>
<div class="container">
<h2>HTTP GET FORM</h2>
<form action="User" method="get">
Name: <input type="text" name="name"><br><br>
Email: <input type="text" name="email"><br><br>
<input type="submit" value="Submit Using Get">
</form>
<br><br>
<h2>HTTP POST FORM</h2>
<form action="User" method="post">
Name: <input type="text" name="name"><br><br>
Email: <input type="text" name="email"><br><br>
<input type="submit" value="Submit Using Post">
</form>
</div>
</body>
</html>
```

User.java:

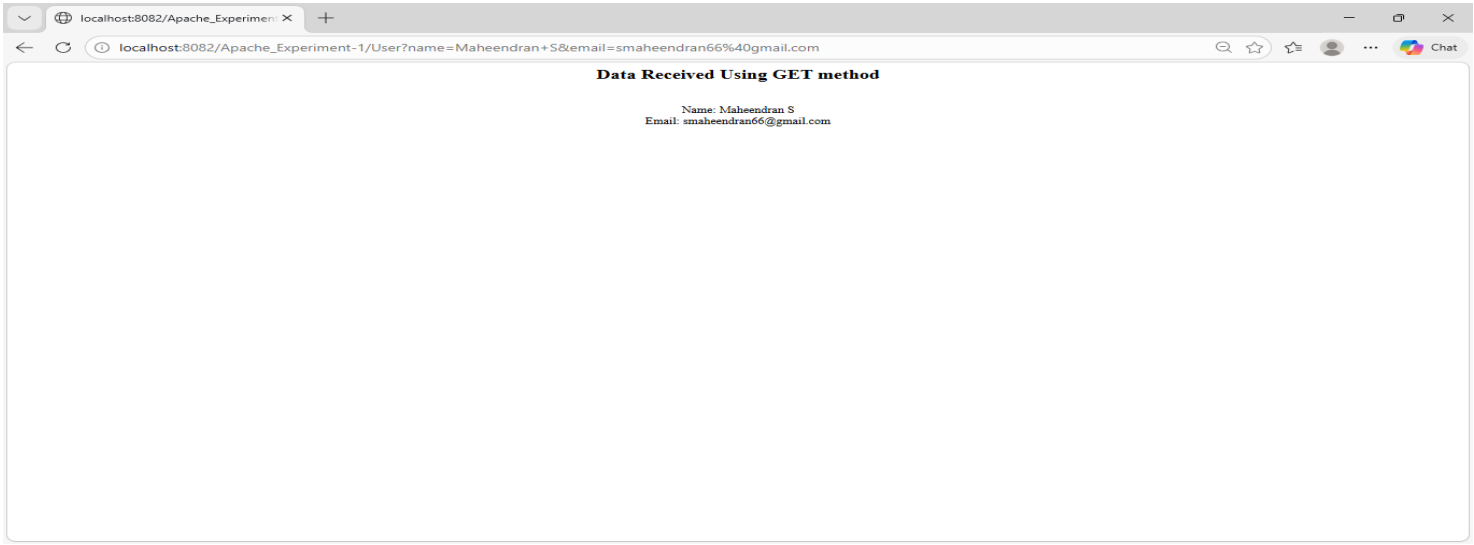
```
package WTMA;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
import java.io.PrintWriter;
@WebServlet("/User")
public class User extends HttpServlet {
        private static final long serialVersionUID = 1L;
    public User() {
        super();

    }
```

```
	protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
		response.setContentType("text/html");
		PrintWriter pw=response.getWriter();
		String name=request.getParameter("name");
		String email=request.getParameter("email");
		pw.println("<center><h2>Data Received Using GET method</h2><br></center>");
		pw.println("<center>Name: "+name+"<br></center>");
		pw.println("<center>Email: "+email+"<br></center>");

	}
	protected void doPost(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
		response.setContentType("text/html");
		PrintWriter pw=response.getWriter();
		String name=request.getParameter("name");
		String email=request.getParameter("email");
		pw.println("<center><h2>Data Received Using POST method</h2><br></center>");
		pw.println("<center>Name: "+name+"<br></center>");
		pw.println("<center>Email: "+email+"<br></center>");


	}

}
```

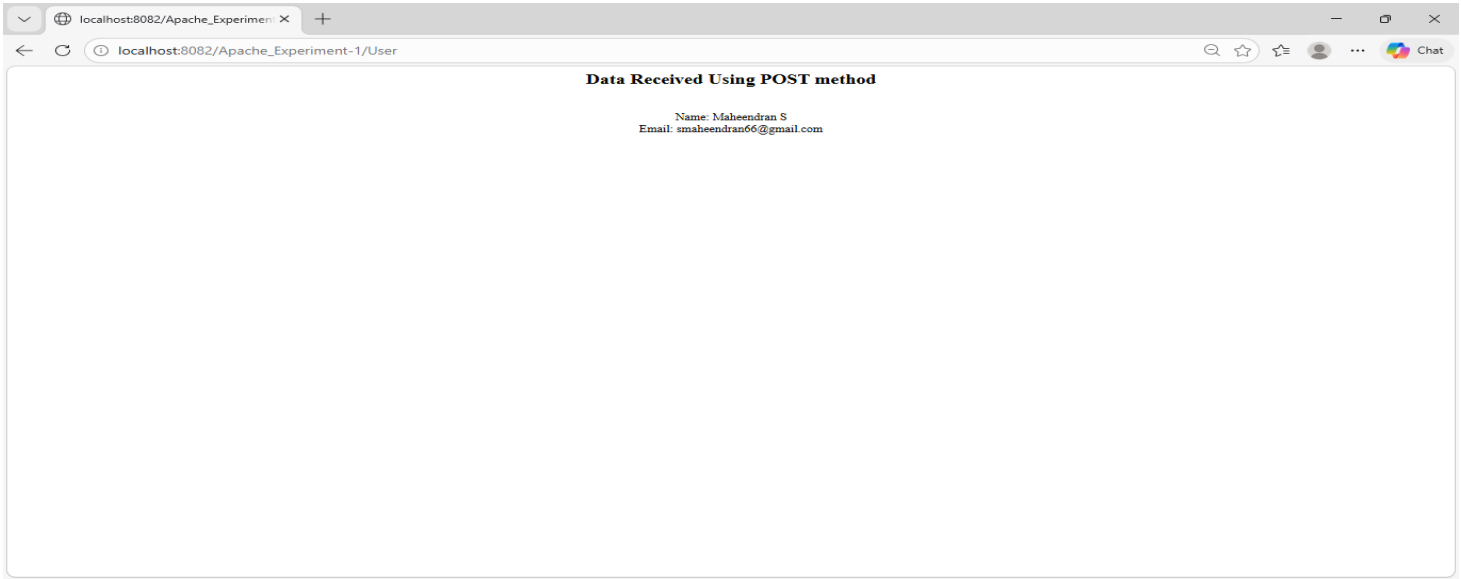
OUTPUT:



OUTPUT-2:



OUTPUT-3:



RESULT:

A Servlet is created to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.